

ADVANCED COMPUTER NETWORKS	
Course Code: CY52	Credits: 2:0:1
Prerequisite: Nil	Contact Hours: 28L+14P
Course Coordinator: Dr. Josy Elsa Varghese	

Course Contents

Unit I

Network layer: Logical addressing - IPV4 addresses, Address space, notations, classful and classless addressing with problem solving, NAT, IPV6 addresses; **Network layer: Internet protocol** - IPV4 datagram, fragmentation, checksum and options; IPV6 packet format, advantages and extension headers; Transition from IPV4 to IPV6.

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Link: https://onlinecourses.nptel.ac.in/noc22_cs19/

Unit II

Address mapping, Error reporting, & Multicasting - Address mapping, ARP, RARP, BOOTP and DHCP; ICMP, IGMP, **Network layer: Delivery, Forwarding, & Routing** – Direct Vs Indirect delivery, Forwarding Techniques, Forwarding Process, Routing Table, **Unicast routing protocols with problem solving** – Optimization, Intra and Inter domain routing, distance vector routing, link state routing.

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Link: https://onlinecourses.nptel.ac.in/noc22_cs19/

Unit III

Multicast routing protocols – Introduction, applications, unicast routing vs multicast routing, source-based tree routing, group shared tree routing. **Transport Layer** - Process-to-Process delivery, User Datagram Protocol, Transmission Control Protocol.

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Link: https://onlinecourses.nptel.ac.in/noc22_cs19/

Unit IV

Congestion control & QOS - Data traffic, Congestion, Congestion control, Two examples – congestion control in TCP and Frame Relay, Quality of Service, Techniques to improve QOS.

Application Layer: Domain Name System - Namespace, Domain name space, Distribution of Name space, DNS in internet, Resolution.

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Link: https://onlinecourses.nptel.ac.in/noc22_cs19/

Unit V

Remote logging – TELNET; Electronic mail – Architecture, User Agent, Message Transfer Agent: SMTP; File transfer - File transfer protocol (FTP),

Network Management: SNMP - Network management system, Simple Network Management Protocol – concept, management components (Basic overview).

- Pedagogy: Chalk and Talk, PowerPoint Presentations
- Link: https://onlinecourses.nptel.ac.in/noc22_cs19/

Text Books:

1. Behrouz A. Forouzan, Data Communications and Networking, Fourth Edition, Tata McGraw-Hill, 2006.

References:

1. Behrouz A. Forouzan, Data Communications and Networking with TCP/IP Protocol Suite| 6th Edition, McGraw-Hill, 2022.
2. Wayne Tomasi, Introduction to Data Communications and Networking, Pearson Education, 2005.
3. Alberto Leon-Garcia and Indra Widjaja, Communication Networks –Fundamental Concepts and Key architectures, Second Edition, Tata McGraw-Hill, 2004.

Course Outcomes (COs):

At the end of the course, students will be able to

1. Understand IPv4 and IPv6 addressing schemes, network layer protocols, and the transition from IPv4 to IPv6.(PO-1, 2, 3,5) (PSO-1, 2)
2. Describe address mapping techniques , error reporting protocols ,packet delivery, forwarding, and routing algorithms.(PO-1, 2, 3,5) (PSO-1, 2)
3. Analyze multicast routing protocols, key transport layer protocols for process-to-process communication.(PO-1, 2, 3,5) (PSO-1, 2)
4. Analyze different techniques to improve transport layer protocols and QOS (PO-1, 2, 3,5) (PSO-1,2)
5. Demonstrate the working of various application layer protocols and network management . (PO-1,2, 3,5) (PSO-1,2)

Course Assessment and Evaluation:

Continuous Internal Evaluation: 50 Marks		
Assessment Tool	Marks	Course outcomes addressed
Internal test-I	30	CO1, CO2, CO3
Internal test-II	30	CO3, CO4, CO5
Average of the two internal tests shall be taken for 30 marks.		
Other components		
Assignment 1	10	CO1, CO2, CO3
Assignment 2	10	CO1, CO2, CO3, CO4, CO5
Semester End Examination (SEE)	100	CO1, CO2, CO3, CO4, CO5